ENS 491/92

FINAL PRESENTATION

GROUP 219

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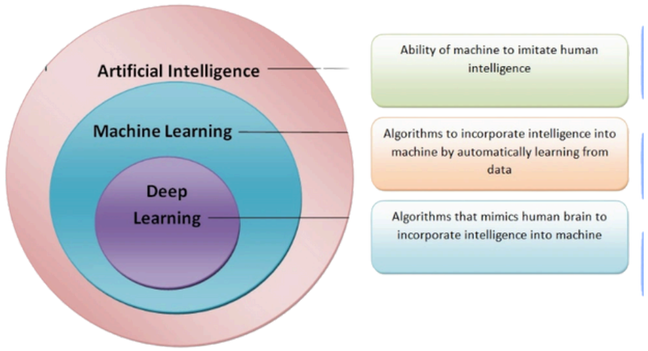
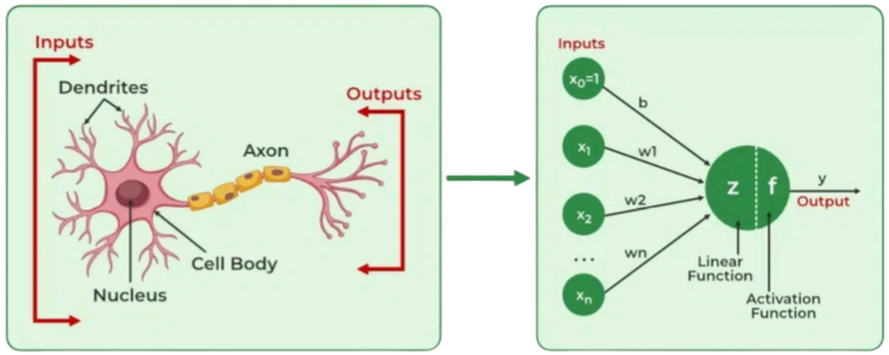
ENS 491
DESIGN

Theoretical Research & Deep Learning Fundamentals

- **Deep Learning Paradigms:** Explored the distinctions between Artificial Intelligence, Machine Learning, and Deep Learning algorithms designed to mimic human brain functions.

- **Historical Evolution:** Analyzed the rise of Deep Learning since 1952, driven by the emergence of Big Data (ImageNet) and parallelizable GPU hardware.

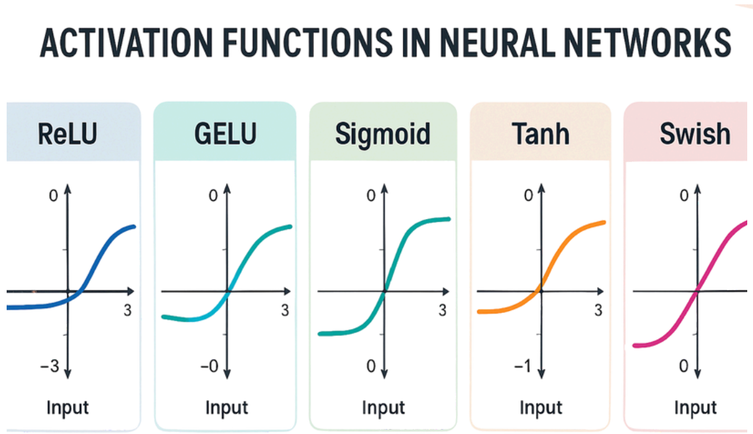


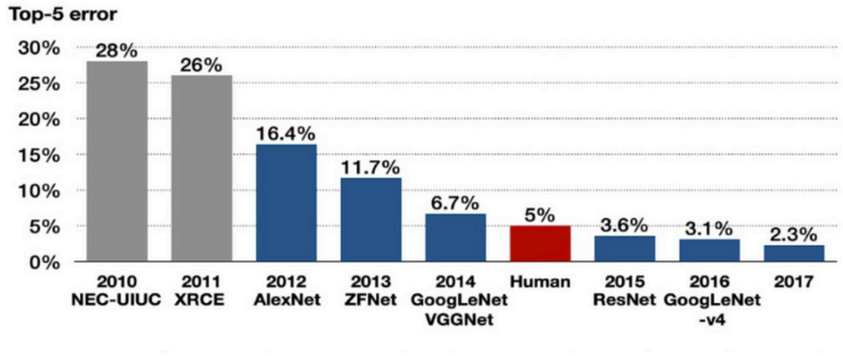


- **Core Components:** Researched neural network essentials including Activation Functions (Sigmoid, Tanh, ReLU), Backpropagation, and Gradient Descent optimization.

- **Benchmarking Datasets:** Evaluated performance metrics using standard datasets such as ImageNet (14M images) and Tiny ImageNet (100,000 images).

- **Overfitting Strategies:** Investigated techniques to improve model generalization, such as Dropout, Data Augmentation, and Early Stopping.

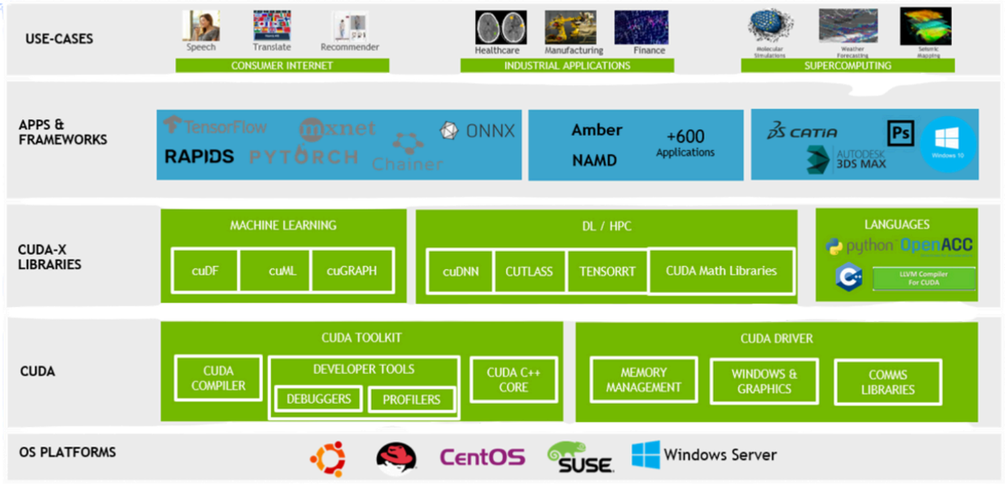




Technical Research & Hardware Optimization

- **CUDA Architecture:** Investigated GPU-accelerated computing using Compute Unified Device Architecture (CUDA) to facilitate massively parallel workloads.

- **Heterogeneous Computing:** Researched models to balance low-latency CPU processing with the high throughput capabilities of GPUs.




```
// CPU vector addition
void vector_add_cpu(float *a, float *b, float *c, int n) {
    for (int i = 0; i < n; i++) {
        c[i] = a[i] + b[i];
    }
}

// CUDA kernel for 1D vector addition
__global__ void vector_add_gpu_1d(float *a, float *b, float *c, int n) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;
    // one add, one multiply, one store
    if (i < n) {
        c[i] = a[i] + b[i];
        // one add, one store
    }
}

// CUDA kernel for 3D vector addition
__global__ void vector_add_gpu_3d(float *a, float *b, float *c, int nx, int ny, int nz) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;
    int j = blockIdx.y * blockDim.y + threadIdx.y;
    int k = blockIdx.z * blockDim.z + threadIdx.z;
    // 3 adds, 3 multiplies, 3 stores

    if (i < nx && j < ny && k < nz) {
        int idx = i + j * nx + k * nx * ny;
        if (idx < nx * ny * nz) {
            c[idx] = a[idx] + b[idx];
        }
    }
```

```
#define N 10000000  // Vector size = 10 million
#define BLOCK_SIZE_1D 1024
#define BLOCK_SIZE_3D_X 16
#define BLOCK_SIZE_3D_Y 8
#define BLOCK_SIZE_3D_Z 8
// 16 * 16 * 8 = 2048
```

```
bash-5.1$ cat vector_add_v2.out
Performing warm-up runs...
Benchmarking CPU implementation...
Benchmarking GPU 1D implementation...
1D Results are correct
Benchmarking GPU 3D implementation...
3D Results are correct
CPU average time: 30.867660 milliseconds
GPU 1D average time: 0.198063 milliseconds
GPU 3D average time: 0.211172 milliseconds
Speedup (CPU vs GPU 1D): 155.847487x
Speedup (CPU vs GPU 3D): 146.172912x
Speedup (GPU 1D vs GPU 3D): 0.937923x
```

- **Lightweight Architectures:** Analyzed the MobileNet family (V1, V2, and V3), focusing on depthwise separable convolutions to reduce parameters for edge device efficiency.

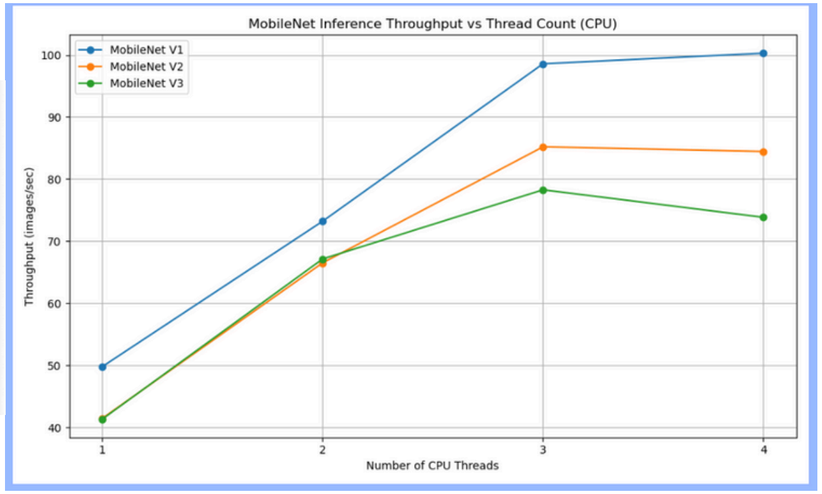
- **Performance Benchmarking:** Conducted initial CPU-based inference tests comparing throughput (images/sec) across different multithreading configurations.

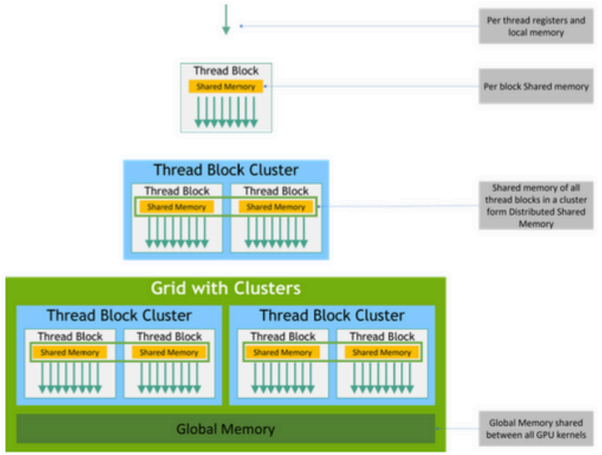
- **GPU Memory Management:** Studied CUDA memory hierarchy, including global, shared, and register memory to optimize data transfer and kernel execution.

MobileNet v2 — logical continuation of the first model, developed in 2018.
It is 0.3x faster and smaller, +1% more accurate than MobileNet v1.

MobileNet v3 is the next generation of the MobileNet family, which is full of improvements to the MobileNet v2, developed in 2019.

It is 2x faster, 30% smaller, -3% accurate than MobileNet v2.





ENS 492 IMPLEMENTATION MOBILENET

High-Performance Computing on TRUBA

- **Infrastructure:** Utilized the Turkish National Science e-Infrastructure (TRUBA) for large-scale training and inference.
- **HPC Environments:** Configured workloads across different node types (Arf, Orfoz, Hamsi) with varying GPU and memory configurations.
- **Job Management:** Implemented sbatch scripts for automated scheduling and resource allocation on SLURM clusters.
- **Storage Optimization:** Leveraged the WEKA high-performance file system (/arf) to minimize I/O bottlenecks during data-intensive tasks.
- **Resource Scaling:** Tested performance across multi-CPU and GPU configurations to determine the optimal compute balance.

TRUBA
Türk Ulusal Bilim e Altyapısı

MobileNet Architecture & Efficiency Analysis

- **Computational Factorization:** Analyzed the shift from standard convolutions to Depthwise Separable Convolutions to reduce Mult-Adds.
- **Mathematical Modeling:** Calculated the theoretical speedup by comparing the costs of standard vs. separable operations.
- **Standard Benchmarking:** Evaluated MobileNet against popular architectures like GoogleNet and VGG16 in terms of parameter counts and ImageNet accuracy.
- **Resolution Multipliers:** Investigated the impact of varying input resolutions (e.g., 224 to 128) on total GFLOPs.

$$\text{Computational Gain} = \frac{\text{Computational Cost of MobileNet}}{\text{Computational Cost of Standard Conv}}$$

$$\text{Computational Gain} = \frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2}$$

Table 6. MobileNet Width Multiplier

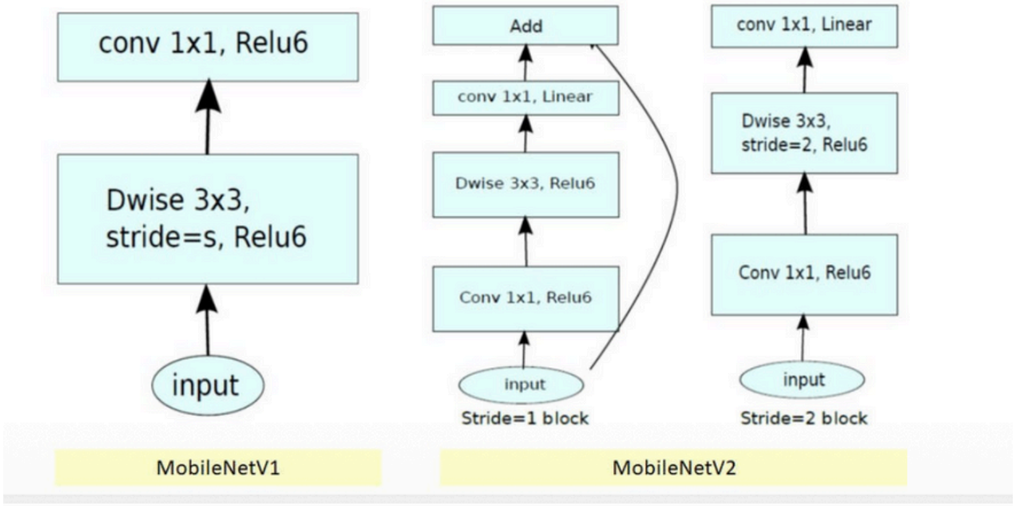
Width Multiplier	ImageNet Accuracy	Million Mult-Adds	Million Parameters
1.0 MobileNet-224	70.6%	569	4.2
0.75 MobileNet-224	68.4%	325	2.6
0.5 MobileNet-224	63.7%	149	1.3
0.25 MobileNet-224	50.6%	41	0.5

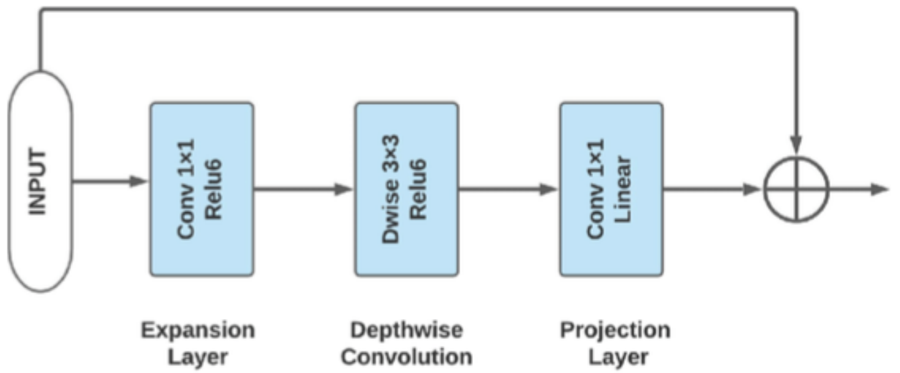
Table 7. MobileNet Resolution

Resolution	ImageNet Accuracy	Million Mult-Adds	Million Parameters
1.0 MobileNet-224	70.6%	569	4.2
1.0 MobileNet-192	69.1%	418	4.2
1.0 MobileNet-160	67.2%	290	4.2
1.0 MobileNet-128	64.4%	186	4.2

MobileNetV2: Inverted Residuals Implementation

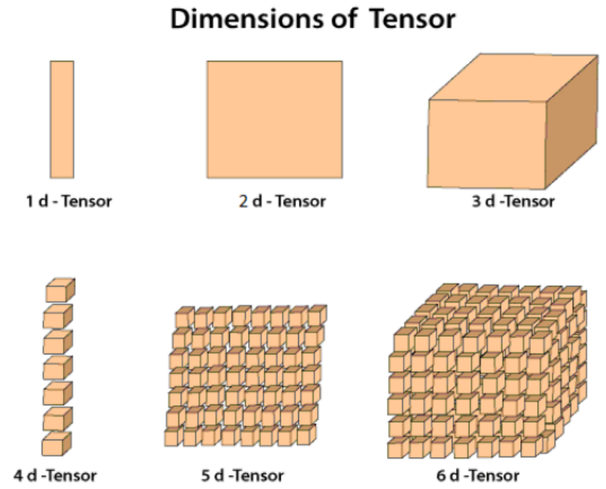
- **Bottleneck Architecture:** Implemented the "Expand-Conv-Squeeze" structure with Linear Bottlenecks to preserve information in low dimensions.
- **Shortcut Connections:** Integrated residual links between narrow layers to facilitate gradient flow and improve training stability.
- **Deployment Mapping:** Mapped the architectural structure to custom tensor formats for later migration to low-level languages.
- **Inference Logic:** Defined the forward pass logic, ensuring proper handling of stride-2 blocks for spatial downsampling.





C++ Forward Pipeline & Tensor Design

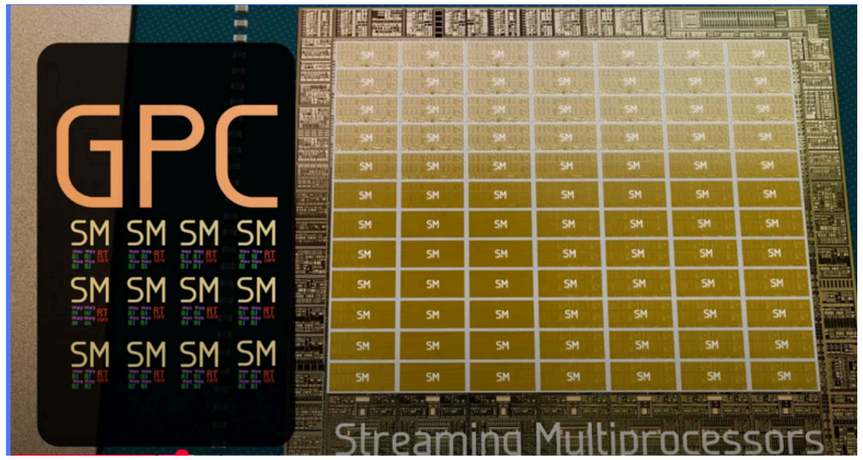
- **Manual Inference Engine:** Developed a C++ system for inference to bypass high-level framework overhead.
- **Tensor Structure:** Designed a custom 3D Tensor class in C++ to manage Channel, Height, and Width (CHW) data layouts.
- **Weight Extraction:** Automated the process of extracting pretrained weights from ONNX models into raw binary format for C++ loading.
- **Validation:** Verified implementation accuracy by comparing C++ output logits with PyTorch reference results on 50,000 images.




```
Images: 50000
Top 1 accuracy: 0.084
Top 5 accuracy: 0.59
Total time: 287.024 sec
Throughput: 174.201 images per sec
Average latency per image: 3.51641 ms
```

GPU Architecture & Scaling Benchmarks

- **Hardware Profiling:** Studied the hierarchy of GPU compute units, including Graphics Processing Clusters (GPCs) and Streaming Multiprocessors (SMs).
- **Warp Execution:** Analyzed the impact of warp divergence on kernel performance, specifically for conditional branching in convolutions.
- **Batch Size Scaling:** Measured the performance of MobileNet across batch sizes ranging from 16 to 1024 to find the GPU saturation point.
- **Metric Analysis:** Evaluated average image time (ms) vs. throughput (img/sec) to identify latency-throughput trade-offs.



📊 === Batch Size GPU Benchmark Results ===

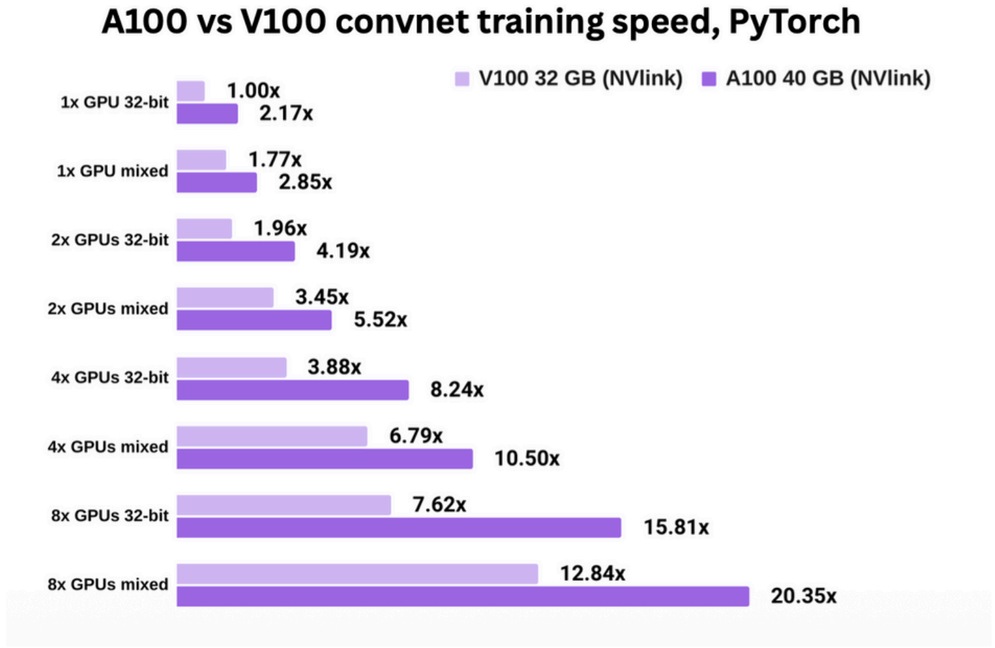
Batch Size	Total Images	Total Time (s)	Avg Batch Time (s)	Avg Image Time (ms)	Throughput (img/sec)	Top-1 Acc (%)	Top-5 Acc (%)
16	50000	303.55	0.0971	6.07	164.7	69.06	88.52
32	50000	160.89	0.1029	3.22	310.8	69.06	88.52
64	50000	96.20	0.1230	1.92	519.7	69.06	88.52
128	50000	59.68	0.1526	1.19	837.8	69.06	88.52
256	50000	46.04	0.2349	0.92	1086.0	69.06	88.52

📊 === Batch Size Benchmark Results ===

Batch Size	Total Images	Total Time (s)	Avg Batch Time (s)	Avg Image Time (ms)	Throughput (img/sec)	Top-1 Acc (%)	Top-5 Acc (%)
16	50000	1239.64	0.3967	24.79	40.3	69.06	88.52
32	50000	888.11	0.5682	17.76	56.3	69.06	88.52
64	50000	886.90	1.1341	17.74	56.4	69.06	88.52
128	50000	838.35	2.1441	16.77	59.6	69.06	88.52
256	50000	1211.01	6.1786	24.22	41.3	69.06	88.52
512	50000	1010.06	10.3067	20.20	49.5	69.06	88.52

Cloud Computing Performance (A100 vs. V100)

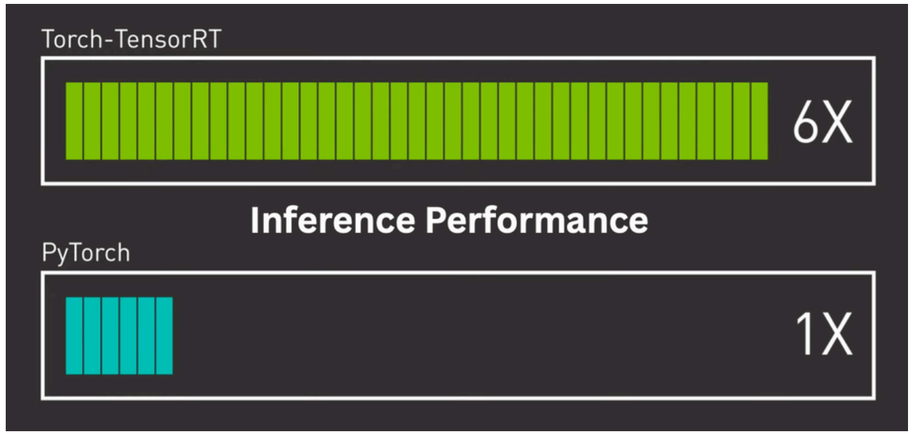
- **High-End GPU Comparison:** Benchmarked inference performance on NVIDIA A100 (Google Colab) vs. V100 (Akya CUDA) architectures.
- **Mixed Precision:** Investigated the throughput gains provided by Tensor Cores when operating in FP16 precision.
- **Inference Frameworks:** Compared standard PyTorch execution against optimized ONNX Runtime and TensorRT workflows on Ampere architecture.

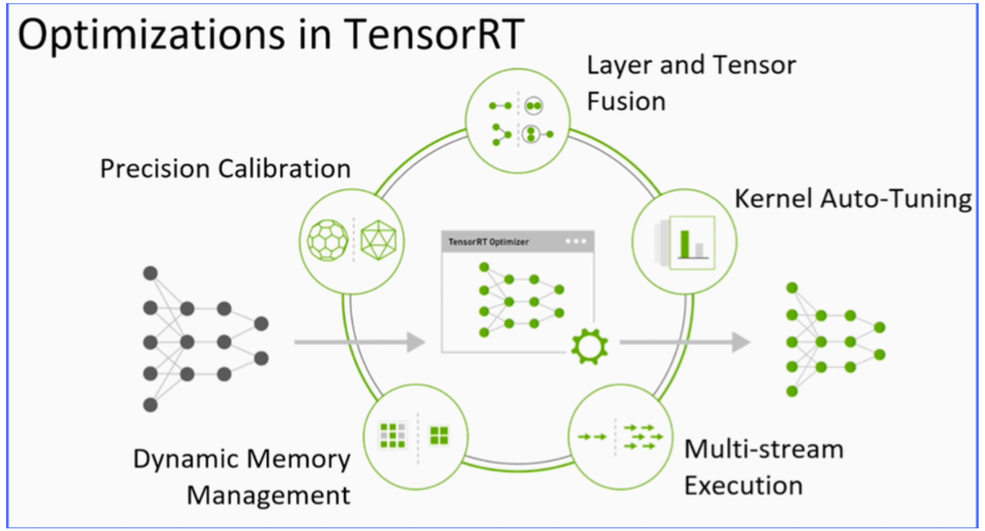



```
PyTorch Latency: 27.914714336395264 ms
PyTorch FPS: 35.82340080393364
ONNX Runtime Latency: 2.9714703559875 49 ms
ONNX Runtime FPS: 336.5337291637414
TensorRT FP16 Latency: 0.38864803314208984 ms
TensorRT FP16 FPS: 2573.022155587237
```

TensorRT Optimization Workflow

- **Optimization Pipeline:** Established a conversion path from PyTorch → ONNX → TensorRT Engine for production-grade inference.
- **Layer Fusion:** Utilized TensorRT's vertical and horizontal layer fusion to reduce memory bandwidth requirements.
- **Kernel Auto-tuning:** Allowed the engine to select the best CUDA kernels for specific GPU architectures (e.g., SM80 for A100).
- **Performance Gains:** Achieved significant throughput increases over standard GPU inference.

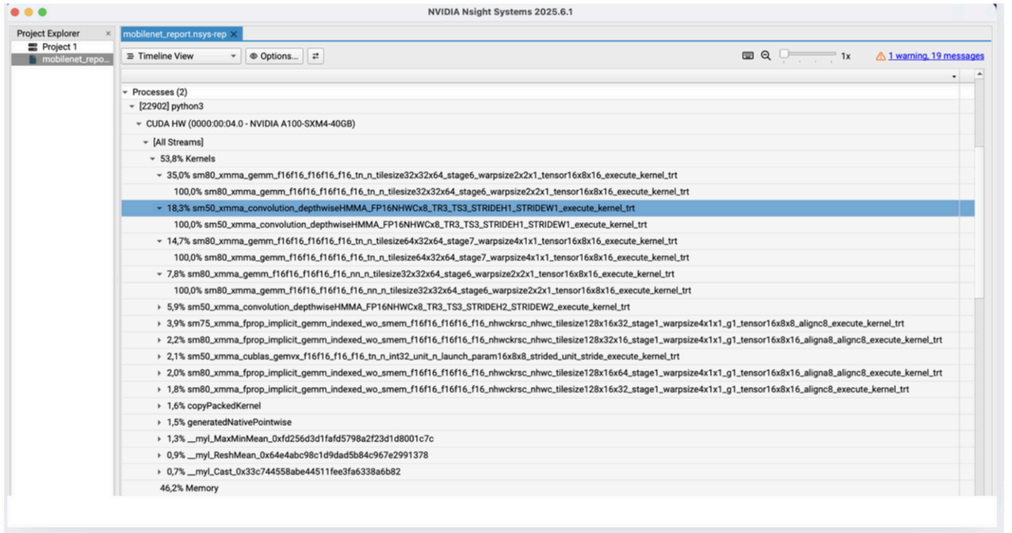




Custom CUDA Kernel Design & Profiling

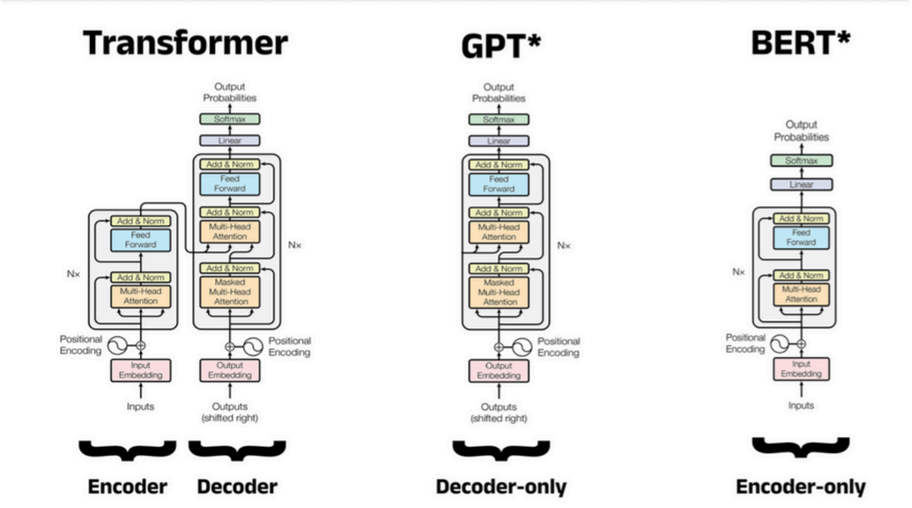
- **Custom Depthwise Kernel:** Developed a specialized FP16 CUDA kernel for 3x3 depthwise convolutions to outperform generic PyTorch kernels.
- **Design Choices:** Simplified kernels to support fixed NCHW layouts, removing unnecessary branching to improve thread utilization.
- **Profiling with Nsight Compute:** Analyzed compute-bound vs. memory-bound bottlenecks using NVIDIA Nsight tools.
- **Kernel Fusion:** Investigated the benefits of fusing activation functions (ReLU6) directly into the convolution kernel.

```
PyTorch FP16 depthwise conv: 0.018766 ms

Custom CUDA kernel: 0.009842 ms

Speedup (PyTorch / Custom): 1.91×
```



Large Language Model (BERT) Optimization

- **Cross-Domain Application:** Applied MobileNet optimization principles to Large Language Models (LLMs) like BERT.
- **Sequence Length Impact:** Measured the effect of padding and sequence length (SEQ_LEN) on inference latency.
- **Backend Comparison:** Tested PyTorch FP16 vs. ONNX Runtime vs. TensorRT for NLP tasks.
- **Throughput Scaling:** Demonstrated how TensorRT provides extreme speedups (~11x) for transformer-based architectures.



BERT ON A100

```
PyTorch FP16 latency: 8.156 ms | Throughput: 122.6 seq/s
ONNX Runtime CUDA latency: 4.123 ms | Throughput: 242.5 seq/s
TensorRT FP16 latency: 0.700 ms | Throughput: 1428.7 seq/s
```

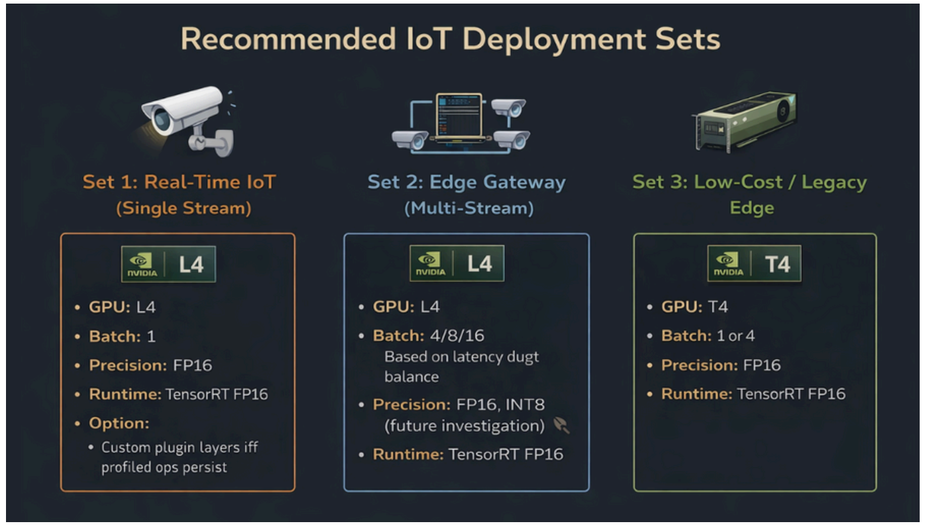
MOBILENET ON A100

```
PyTorch Latency: 22.01592445373535 ms
PyTorch FPS: 45.42166748897679
ONNX Runtime Latency: 2.5336303709375 ms
ONNX Runtime FPS: 394.6905639469056
TensorRT FP16 Latency: 0.34925031661987305 ms
TensorRT FP16 FPS: 2863.275858067176
```

Performance Standards for IoT Deployment

- **IoT Latency Targets:** Researched real-time vision requirements, targeting <33ms (30 FPS) for end-to-end processing.
- **Watt-per-Performance:** Evaluated the NVIDIA L4 GPU for edge scenarios where power efficiency is as critical as throughput.
- **Precision Recommendations:** Concluded that FP16 with TensorRT + Custom Kernels is the optimal configuration for modern IoT gateways.
- **Use-Case Scenarios:** Defined deployment sets for real-time single-stream vs. multi-stream edge gateways.

Bileşen	Seçim	Gerekçe
GPU	NVIDIA L4	A100 daha hızlı olsa da, L4 özellikle edge veya inference yoğun iş yükleri için tasarlanmıştır ve watt başına yüksek performans sunar.
Batch Size	1	IoT senaryolarında yanıt süresini, yani latency'yi en aza indirmek için kritiktir. Sonuçlarımıza göre, L4 ve A100 üzerinde Batch 1 için 0.23 ms ile 0.34 ms arasında latency göstermektedir.
Precision	FP16	Test edilen tüm GPU'larda latency ve throughput açısından FP32'ye kıyasla tutarlı biçimde daha iyi performans göstermektedir, hızlanma bazı durumlarda 2 kata kadar çıkmaktadır.
Engine	TensorRT + Custom Kernel Plugin	Özel depthwise kernelı 2.07 kat, pointwise kernelı ise 2.61 kat hızlanma sağlayarak standart PyTorch FP16 uygulamasını belirgin şekilde geride bırakmaktadır.



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THANK YOU!